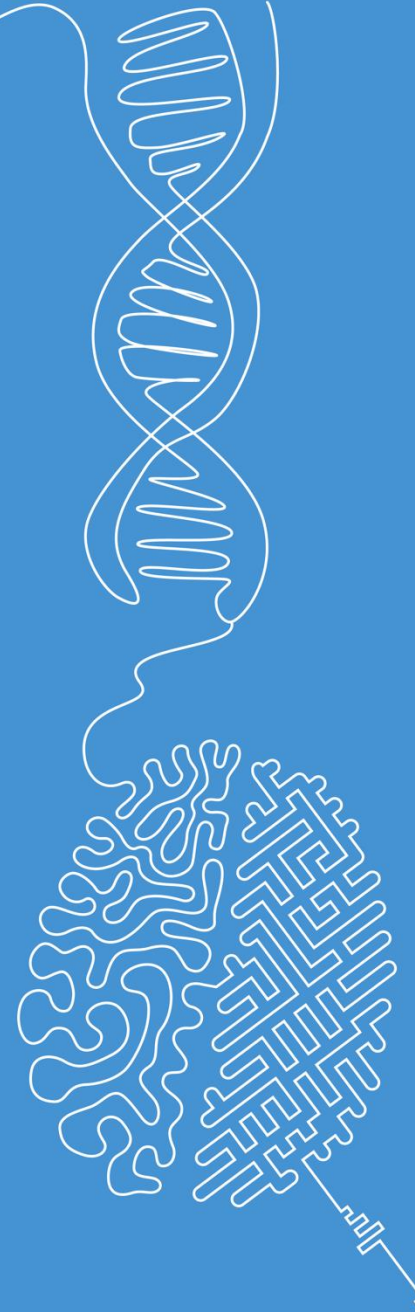


Fourier Transformation

Image and Signal Processing

Norman Juchler



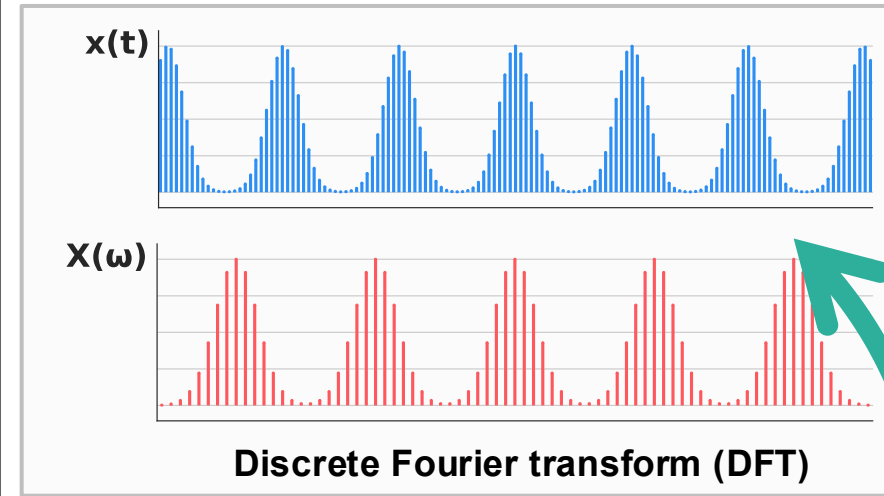
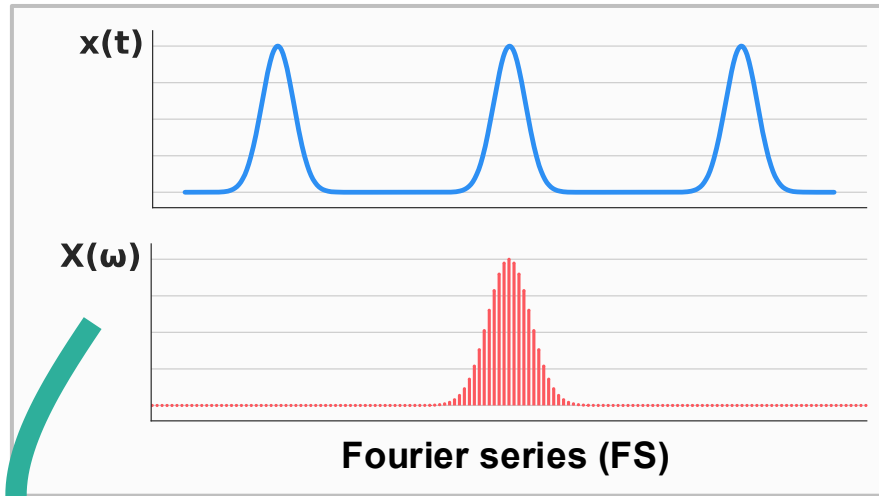
We began our journey into the frequency domain with Fourier series for periodic signals. We now generalize this idea to aperiodic signals and then continue to its discrete counterparts.

Fourier's landscape

Time-continuous

Time-discrete

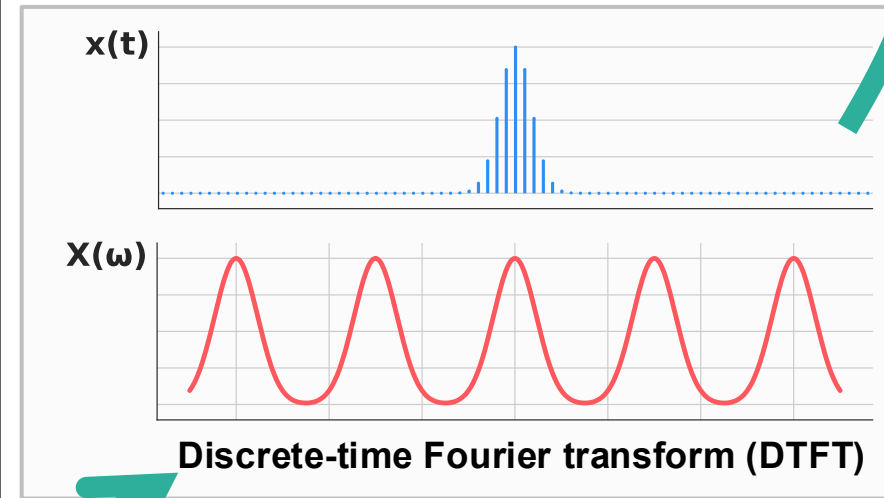
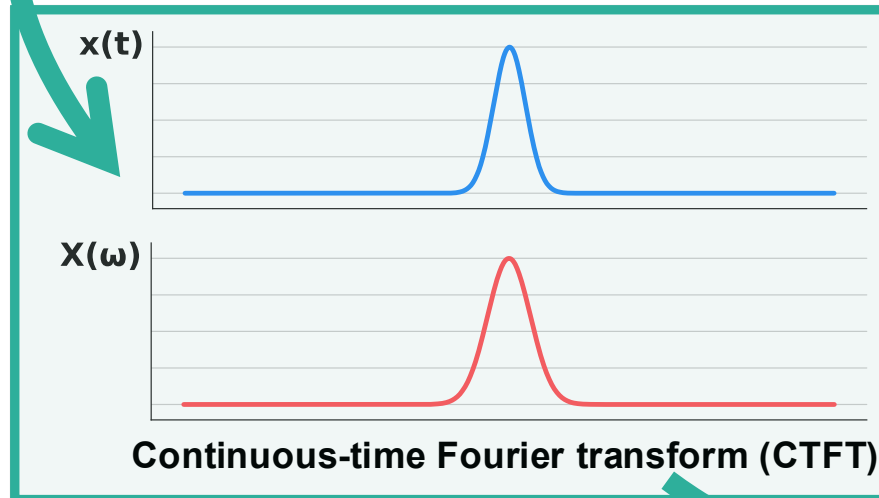
Periodic



Related:

- Fast Fourier transform
- Discrete cosine transform
- Discrete sine transform

Aperiodic



Generalization:
z-Transform

Generalization:
Laplace
Transform

Topic of today!

Fourier transformation – Outline

- **Idea:** Treat the aperiodic function $x(t)$ like a periodic function with period $T \rightarrow \infty$ and then compute Fourier series. In the limit, this results in the Fourier transform.

- **Fourier transform**

$$\hat{x}(k) = \int_{-\infty}^{\infty} x(t) e^{-ikt} dt \quad k \in \mathbb{R}$$

- **Inverse Fourier transform**

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{x}(k) e^{ikt} dk$$

For comparison: the Fourier series formulas

$$c_k = \frac{1}{T} \int_0^T x(t) \cdot e^{-i2\pi \frac{k}{T} t} dt$$

$\forall k \in \mathbb{Z}$

$$x(t) = \sum_{k=-N}^N c_k e^{\frac{2\pi i k}{T} t}$$

...do you see similarities?

- The Fourier transform generalizes the Fourier series to aperiodic signals. It allows us to analyze signals that do not repeat periodically.
- Disclaimer: The above integrals exists for many, but not all, signals. Here, we skip the technical details.

Immediate observations

- With the **Fourier transformation**, we bring a continuous-time signal into the frequency domain
- With the **inverse Fourier transformation**, we bring a frequency domain signal back into the time domain
- Alternative notations:
 - Exponential form:

$$\hat{x}(\omega) = X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-i\omega t} dt$$

- Sine-cosine form:

$$\hat{x}(\omega) = X(\omega) = \underbrace{\int_{-\infty}^{\infty} x(t) \cos(\omega t) dt}_{R(\omega t)} - i \underbrace{\int_{-\infty}^{\infty} x(t) \sin(\omega t) dt}_{-I(\omega t)}$$

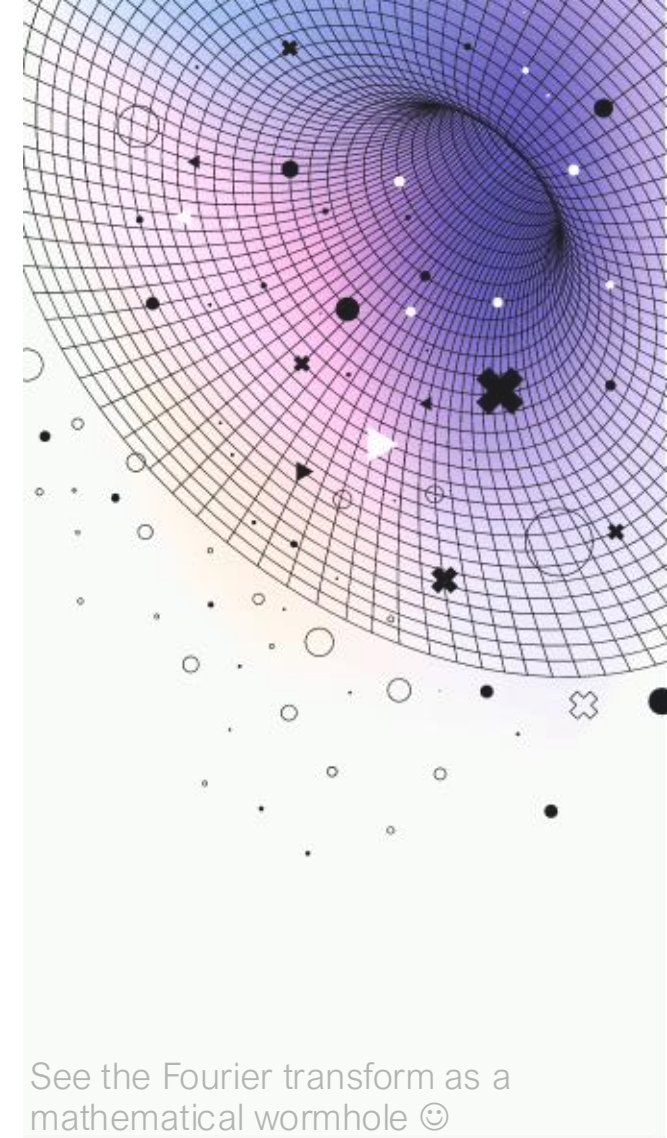
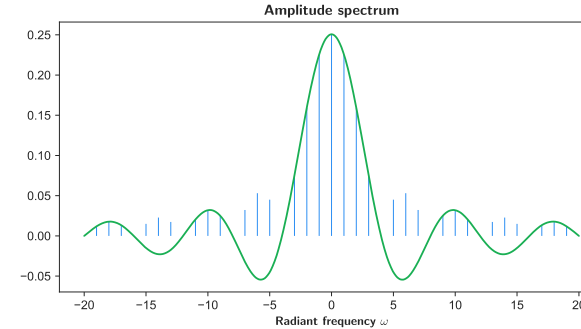
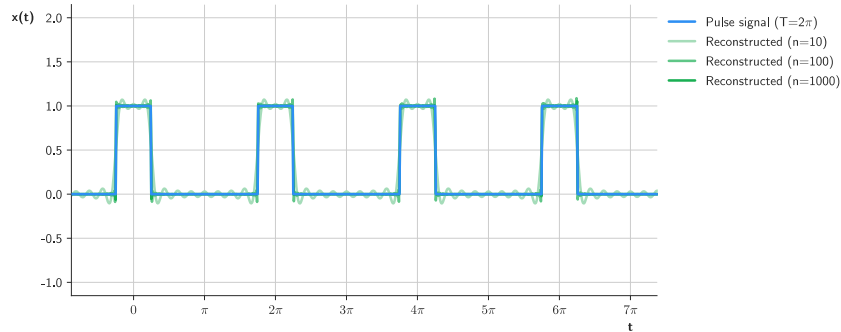
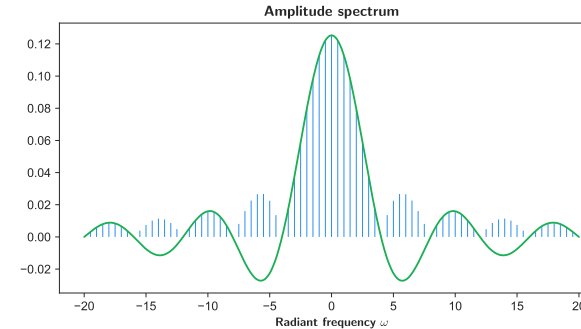
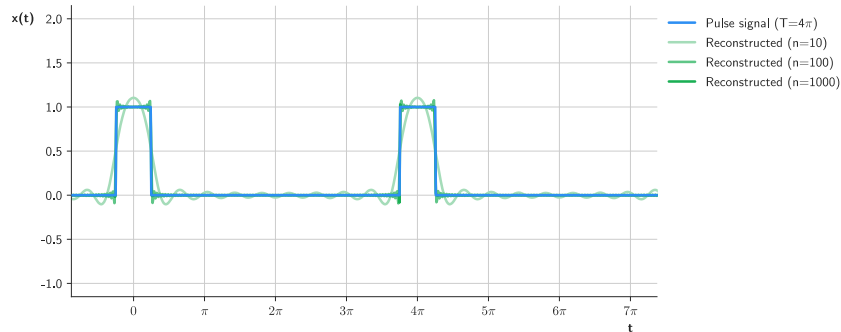


Illustration: Fourier series of the pulse wave for increasing T

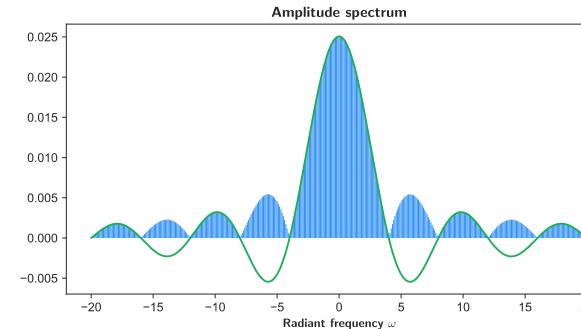
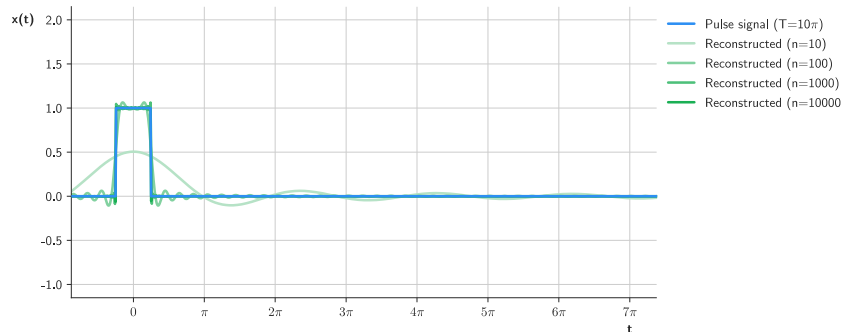
$$\begin{aligned} A &= 1 \\ T &= 2\pi \\ r &= \frac{1}{4} \end{aligned}$$



$$\begin{aligned} A &= 1 \\ T &= 2 \cdot 2\pi \\ r &= \frac{1}{4} / 2 \end{aligned}$$



$$\begin{aligned} A &= 1 \\ T &= 10 \cdot 2\pi \\ r &= \frac{1}{4} / 10 \end{aligned}$$



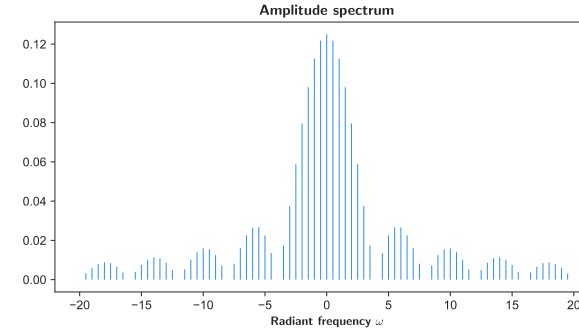
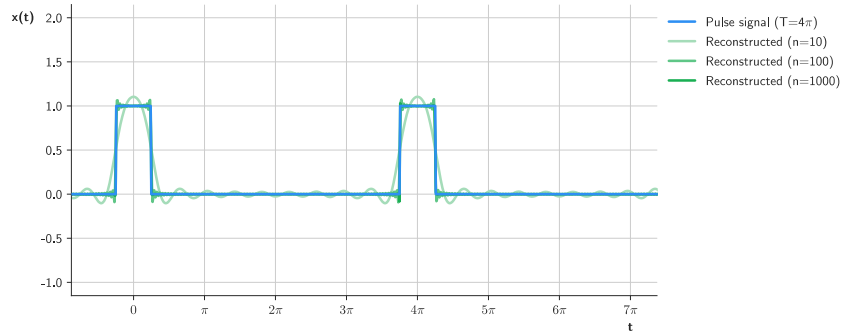
Observations:

- The larger the period, the denser the samples in the frequency domain
- The envelope is linked to the **Fourier transform** of the aperiodic pulse at $t = 0$.
- The discrete series coefficients form a continuity in the extreme case $t \rightarrow \infty$

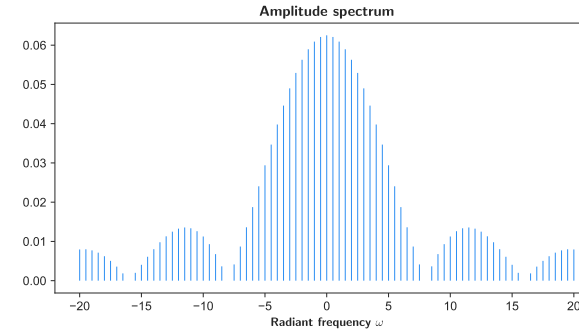
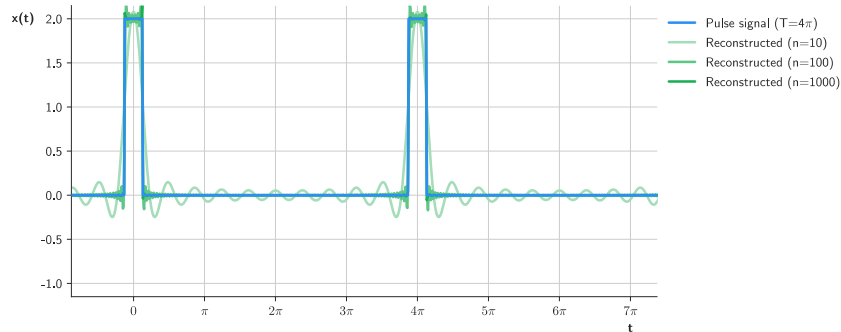
$$x(t) = \begin{cases} A, & |t| \leq \frac{r \cdot T}{2} \\ 0, & |t| > \frac{r \cdot T}{2} \end{cases}$$

Illustration: Fourier series of the pulse wave for decreasing width

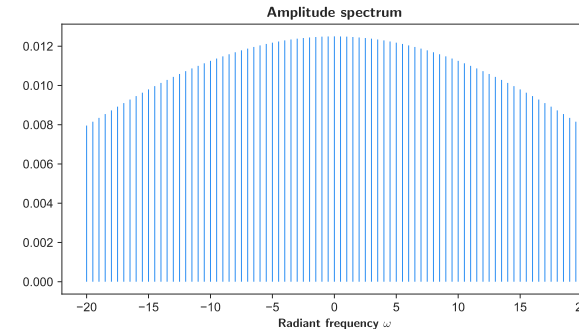
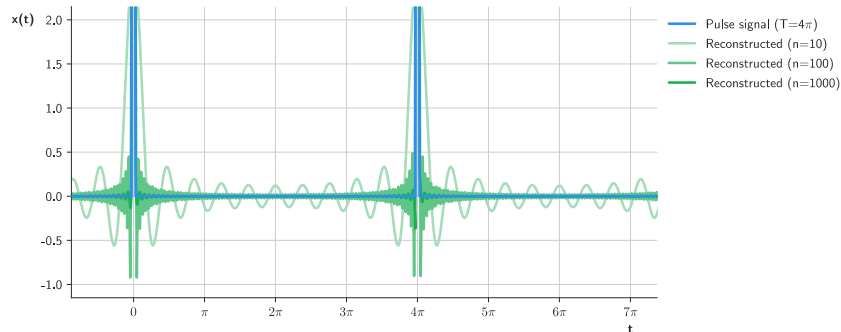
$$\begin{aligned} A &= 2 \\ T &= 2 \cdot 2\pi \\ r &= \frac{1}{4} / 2 \end{aligned}$$



$$\begin{aligned} A &= 4 \\ T &= 2 \cdot 2\pi \\ r &= \frac{1}{4} / 4 \end{aligned}$$



$$\begin{aligned} A &= 10 \\ T &= 2 \cdot 2\pi \\ r &= \frac{1}{4} / 10 \end{aligned}$$



Observations:

- The narrower the pulses in the time-domain, the wider they are in the frequency domain
- In the extreme case, with infinitely thin pulses, the spectral coefficients are constant.

Example: Rectangular function

$$x(t) = \begin{cases} A, & -\tau/2 \leq t \leq \tau/2 \\ 0, & \text{otherwise} \end{cases}$$

$$\hat{x}(\omega) = \int_{-\infty}^{+\infty} x(t) e^{-i\omega t} dt, \quad \omega \in \mathbb{R}.$$

Derivation:

$$\begin{aligned} \hat{x}(\omega) &= \int_{-\tau/2}^{+\tau/2} A e^{-i\omega t} dt = \int_{-\tau/2}^{+\tau/2} A \cos(\omega t) dt - i \int_{-\tau/2}^{+\tau/2} A \sin(\omega t) dt \\ &= \frac{A}{\omega} \sin(\omega t) \Big|_{-\tau/2}^{+\tau/2} = \frac{2A}{\omega} \sin\left(\frac{\tau}{2}\omega\right) = \frac{2A}{2\pi f} \sin\left(\frac{\tau}{2}2\pi f\right) \\ &= \frac{A\tau}{\tau\pi f} \sin(\tau\pi f) = \underline{\underline{A\tau \operatorname{sinc}(\tau f)}} \end{aligned}$$

Use here that $\omega = 2\pi f$

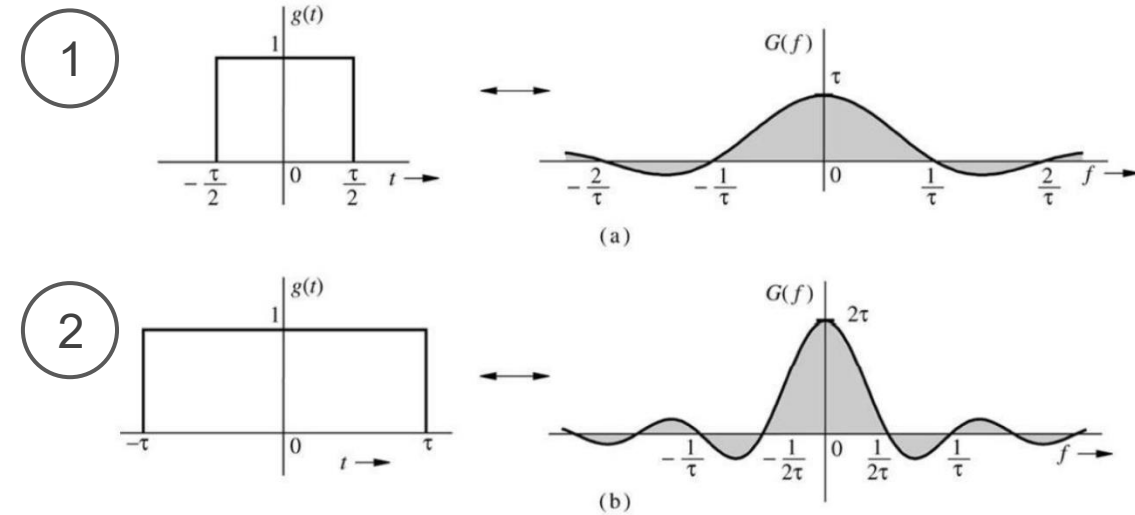


Figure 1: A rectangular function of width τ and its Fourier transform, a sinc function compressed horizontally by a factor of $1/\tau$.

Figure 2: A rectangular function with doubled width 2τ . The broader time-domain signal results in a narrower Fourier transform — doubling the width in time halves the width in frequency.

Example: Impulse function

■ Idea:

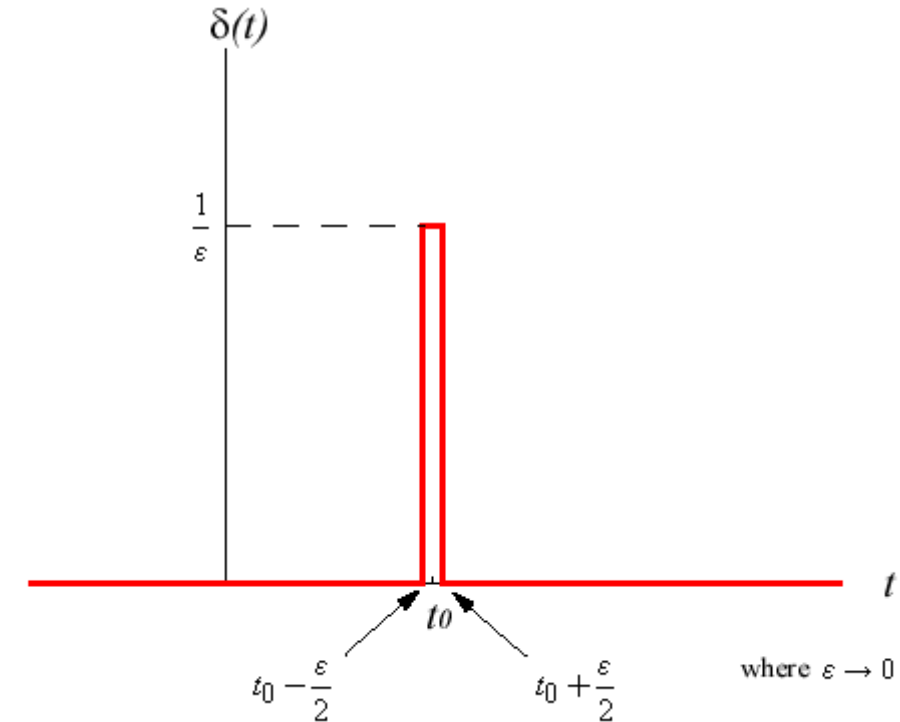
- Begin with a rectangular pulse of width 1 and area 1
- Reduce the width, while keeping the area
- Take this process to the extreme: $\epsilon \rightarrow 0$
- Such a function is called impulse function $\delta(t - t_0)$

■ Compute the Fourier transform of

$$\hat{x}(\omega) = \int_{-\infty}^{+\infty} x(t) e^{-i\omega t} dt, \quad \omega \in \mathbb{R}.$$

$$\hat{x}(\omega) = \int_{-\infty}^{+\infty} \delta(t - t_0) e^{-i\omega t} dt = 1$$

Observe that the Fourier transform of an impulse (Dirac delta) is constant. You can think of the impulse as an extreme case of a rectangular pulse: as the pulse becomes infinitely narrow in time, its Fourier transform becomes infinitely broad in frequency.

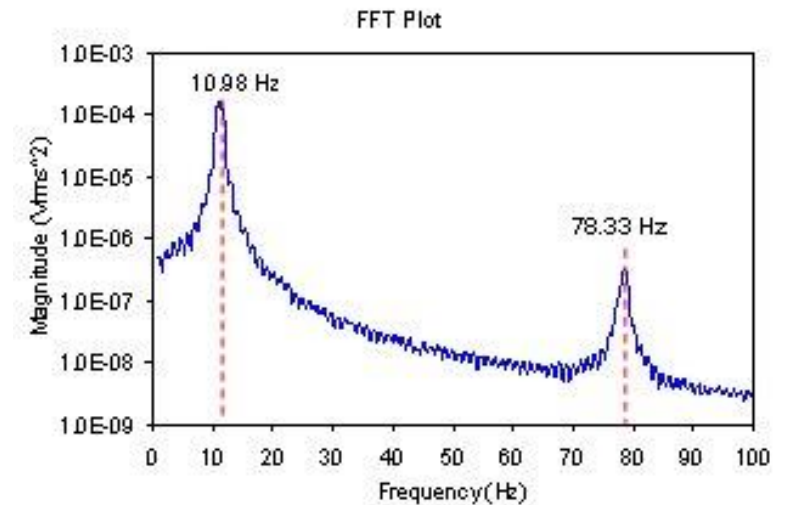
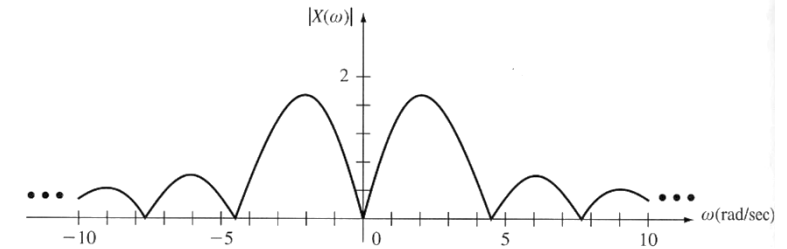


Fourier transform and spectra

- The Fourier transform $X(\omega)$ is generally a continuous, complex-valued function.
- Like any complex number, it can be written in polar form:

$$X(\omega) = |X(\omega)|e^{i\varphi(\omega)}, \quad \varphi(\omega) = \arg(X(\omega))$$

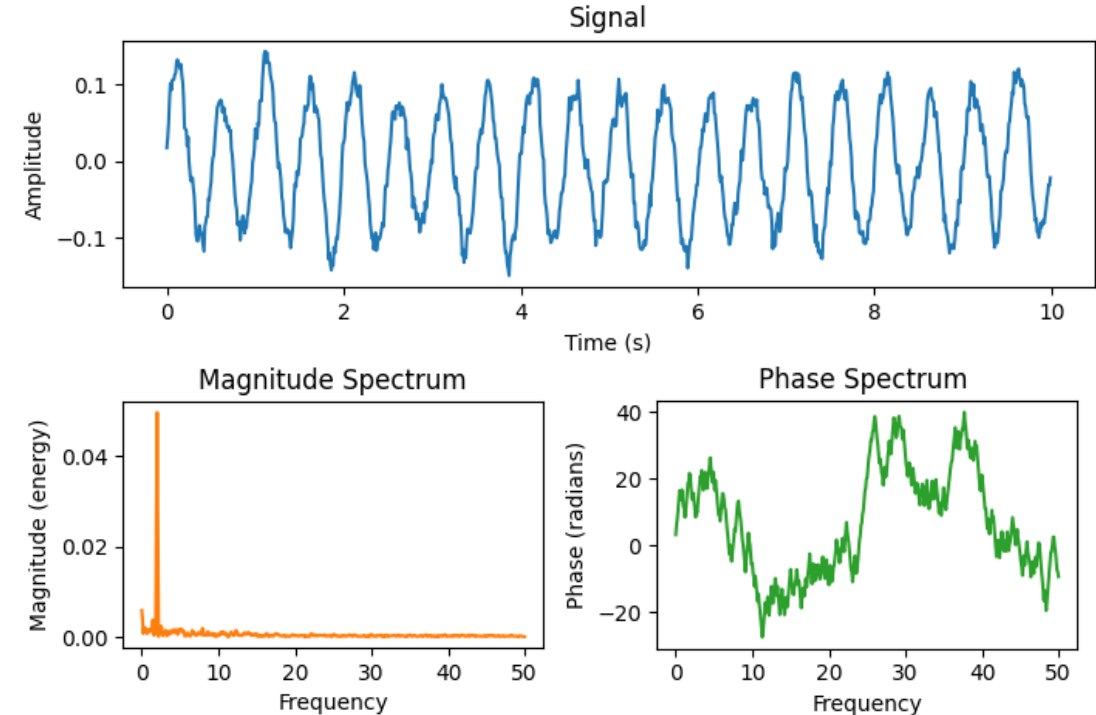
- Both amplitude spectrum $|X(\omega)|$ and the phase spectrum $\varphi(\omega)$ are continuous functions in ω .
- **Note:** Negative frequencies are a “mathematical artifact” of Fourier theory, which is why spectra sometimes extend to negative ω . For real-valued signals $x(t)$, the content of the FT at negative frequencies is redundant.



Examples of an amplitude spectrum of a continuous-time signal (top) and a power spectrum of a system's impulse response (bottom). The impulse response is a system's output when it is excited by an ideal, instantaneous impulse.

The magnitude (or amplitude) spectrum

- A magnitude spectrum plots the magnitude of Fourier coefficients (or Fourier transform) as a function of frequency (f or ω).
- It reveals how strongly each frequency contributes to the signal and describes the distribution of energy.
- It is essential for studying signals and for modifying their frequency components for various purposes (e.g., noise reduction, signal enhancement, or more generally, filtering).
- Important: Signals with the same magnitude spectrum can still differ if their phase differs.



The phase

- A phase spectrum plots the phase φ of Fourier coefficients as a function of frequency (f or ω).
- Analyzing phase across frequencies provides important insights into a signal's structure
- Phase shifts determine **timing**, **alignment**, and **interference patterns**.
- Large or irregular phase shifts can distort the reconstructed signal's shape and timing
- For curious students: A great visual explanation of the role of phase appears in this video: [On holograms \(by 3blue1brown\)](#)

